Goal Selection for Minimum Jerk Wrist Trajectory Approximation

This follow-up compares two adaptive goal-selection strategies with uniform sampling: reconstruction-error-driven selection and motion-change-based selection using acceleration magnitude. At 102 goals, the revised error-driven method achieved the lowest reconstruction errors. I also compare representative goal counts to examine how additional goals change position and velocity reconstruction; a final count is not selected automatically.

1 Motivation and data

The initial experiment sampled goals every 0.5 seconds. It followed many broad position changes but missed brief peaks and showed timing differences. Following your feedback, I considered how to select goals from the motion rather than from a regular clock. The overlapping minimum jerk displacement formulation was retained; the changes in this experiment concern goal selection and the rule assigning trajectory timing.

The data are unchanged: `ACCAD/Female1Gestures_c3d/D3 - Conversation Gestures_poses.npz` from AMASS, with 6,086 frames at 120 Hz (approximately 50.7 seconds). The female SMPL+H model provides the right-wrist global XYZ positions in meters. The whole clip is used. The evaluation reference is this model-derived wrist trajectory, without smoothing, cropping, or resampling.

2 Two initial ideas

My first idea was to add goals where the current minimum jerk reconstruction differs most from the original trajectory. The reasoning was to spend additional goals on the parts that the current representation fails to describe. This requires repeated reconstruction, which is acceptable for this offline conversion experiment.

My second idea was to place goals where the original motion changes substantially. Unlike the first approach, its candidate locations can be determined before generating an approximation. To turn this idea into a concrete rule, I chose acceleration magnitude as a measure of change in the velocity vector. It does not measure position excursion or the rate of change of acceleration, which would be jerk.

Neither idea alone specifies the activation and completion times, a unique appropriate goal count, or a guarantee of lower error. These choices and limitations are made explicit in the following sections.

3 Shared representation and evaluation

All methods use goal positions sampled from the original trajectory. I did not fit goal positions in this comparison, although that is possible. The initial and final positions are retained. A count of K goals excludes the initial position and includes the final position, so K trajectories connect K+1 source samples.

For source timestamps `t_0 < ... < t_K`, let `g_i` be the corresponding positions and `Delta_i = t_i - t_(i-1)`. At each interior boundary, define:

```
delta_i = 0.25 * min(Delta_i, Delta_(i+1))
delta_0 = delta_K = 0
a_i = t_(i-1) - delta_(i-1)
b_i = t_i + delta_i
```

Here `a_i` and `b_i` are activation and completion times. This allows neighboring instances to overlap while reducing the extension near short source intervals. The factor 0.25 is a fixed initial choice, not an optimized or inferred value. For equal 0.5-second intervals, each interior side extends by 0.125 seconds. Clip endpoints are not extended.

For `u_i = (t - a_i) / (b_i - a_i)`, the progress and reconstruction are:

```
s_i(t) = 0                                 before a_i
         10u_i^3 - 15u_i^4 + 6u_i^5        from a_i to b_i
         1                                 after b_i
p_hat(t) = g_0 + sum_i (g_i - g_(i-1)) * s_i(t)
```

Displacements are summed as in Sato et al. (2025), Section 3.4.1. Source timestamps identify where positions were sampled, not when the combined trajectory must pass through them. During overlap, the output need not reach a sampled goal at its source timestamp or its instance completion time.

All metrics use all N original timestamps. With Euclidean norm denoted by `|| ||`, the definitions are:

```
Position RMSE = sqrt(mean_k ||p_hat_k - p_k||^2)
Velocity RMSE = sqrt(mean_k ||v_hat_k - v_k||^2)
Maximum position error = max_k ||p_hat_k - p_k||
```

Position metrics are reported in centimeters; velocity RMSE is in m/s and includes direction differences. Both velocities use `numpy.gradient` with original timestamps, centered interior differences, and second-order one-sided endpoint differences (`edge_order=2`). The metrics are reported separately, not combined into a weighted score. A common minimum gap of 0.1 seconds applies to selected source timestamps. The gap and overlap are settings for this initial comparison, not estimated properties of human motion.

4 Error driven selection and its revision

Original attempt

I began with the initial and final samples, reconstructed the clip, and selected the eligible frame with the largest 3D position distance from the reconstruction. A candidate is eligible if it has not already been selected and lies at least 0.1 seconds from every selected source timestamp.

The first candidate was frame 5038 at 41.983333 seconds. Adding it increased position RMSE from 36.589 cm to 44.024 cm. This initially seemed counterintuitive: why would adding a point where the error is large make the overall result worse?

The insertion does more than correct that instant. It splits the original displacement into two displacements and recomputes their timing. With so few goals, an instance lasts for tens of seconds and changes positions well outside the short peak it was intended to capture. The first-insertion plot shows a broad reconstructed rise in Z instead of a brief peak. Therefore,

retaining more source positions does not preserve the previous approximation or guarantee lower error.

The unconditional version was run up to 102 goals. Its final position RMSE was 1.604 cm, but position RMSE rose on seven of the 101 insertions. These results were preserved rather than discarded.

Revised acceptance and rollback rule

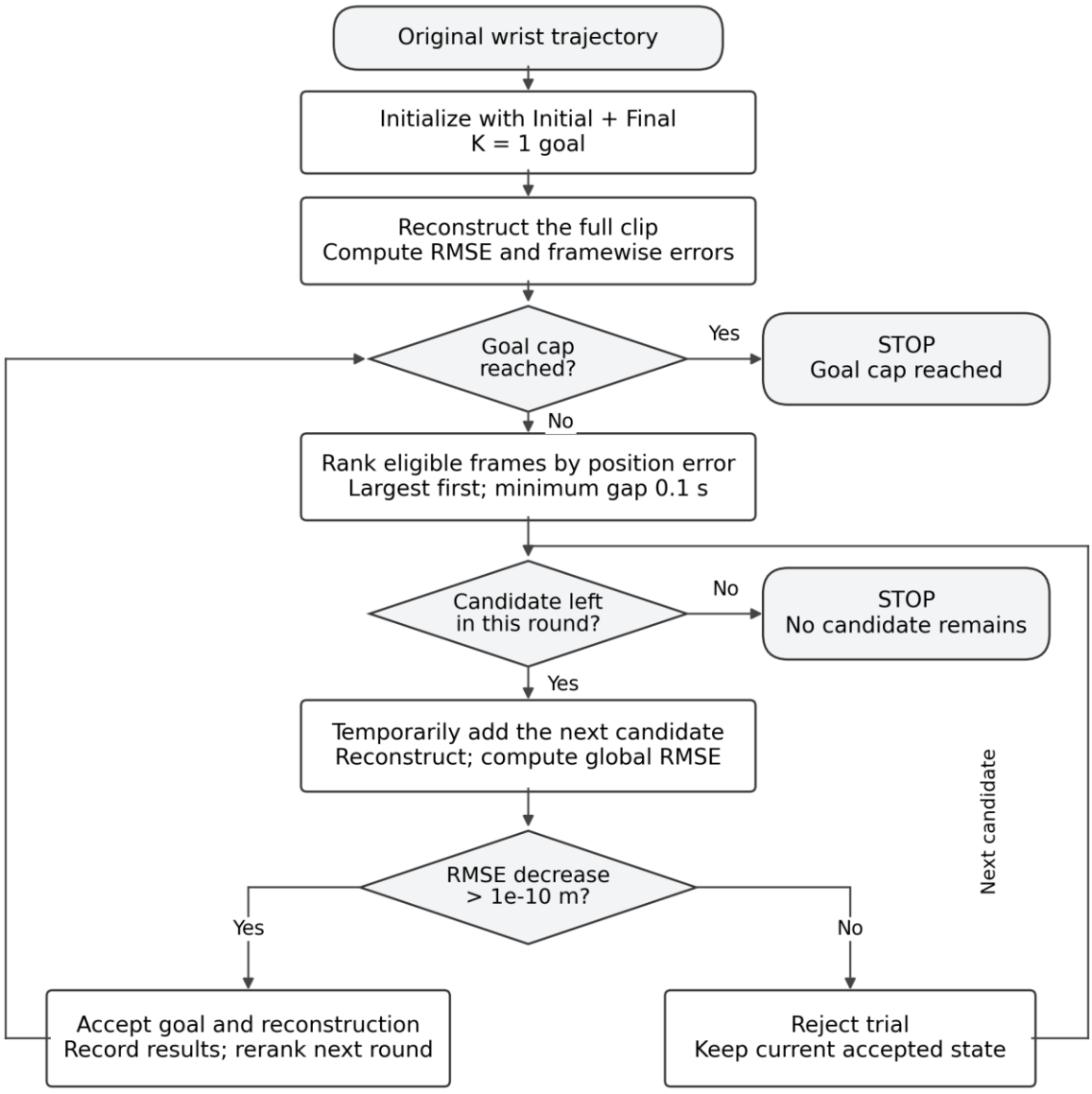
I revised the idea so that a large error makes a point worth trying, not automatically worth accepting:

1. Compute current per-frame 3D position errors and rank eligible candidates from largest to smallest. Ties use the earlier frame.
2. Temporarily insert the highest-ranked remaining candidate, take its original XYZ as the goal, and regenerate the complete approximation with updated timing.
3. Accept the first candidate that lowers global position RMSE by more than `1e-10 m`. Otherwise discard the trial and try the next candidate against the unchanged accepted state.
4. After acceptance, record the result and recalculate errors and rankings. Previously rejected points may be reconsidered in the next round.
5. Stop at the goal budget, if no eligible frame remains, or if an entire candidate round yields no improvement.

This is first improvement, not a search for the greatest improvement among all candidates. The numerical tolerance avoids accepting negligible changes; it is not a target reconstruction accuracy. Velocity RMSE and maximum position error are recorded but do not control acceptance.

The revised run reached 102 goals after 6,283 trials: 101 accepted and 6,182 rejected. In the first round, the first 29 candidates were rejected and rank 30, frame 5020 at 41.833333 seconds, was accepted. Position RMSE fell slightly to 36.560 cm. At 102 goals it reached 1.478 cm. However, velocity RMSE was 0.17128 m/s versus 0.16457 m/s for unconditional insertion, and maximum position error was 5.846 cm versus 4.402 cm. Enforcing improvement in one aggregate metric did not improve every metric.

4.1 Flowchart of error driven selection



Flowchart 1. First-improvement insertion with acceptance and rollback. Rejected candidates are tested against the unchanged accepted reconstruction. After acceptance, errors and candidate rankings are recomputed. The 1e-10 m tolerance is numerical; it is not a quality threshold. This experiment stopped at the prescribed cap of 102 goals.

5 Selecting goals from motion changes

The intended reasoning was to identify where the original movement changes, rather than where a particular reconstruction is inaccurate. I made that idea measurable with the score:

```
score(t) = ||a(t)||
a(t) = d v(t) / dt
```

Acceleration magnitude responds to both changes in speed and changes in direction. It is only a proxy for the importance of a goal to reconstruction. Large spatial movement does not necessarily imply large acceleration, and a short fluctuation can give a large score without a large position excursion.

Velocity and acceleration are estimated by applying `numpy.gradient` twice to the raw positions with the original timestamps and `edge_order=2`. No smoothing is used. Candidates are positive interior local maxima of the score; an exactly flat peak uses its earliest frame. The endpoints are included separately.

The procedure is:

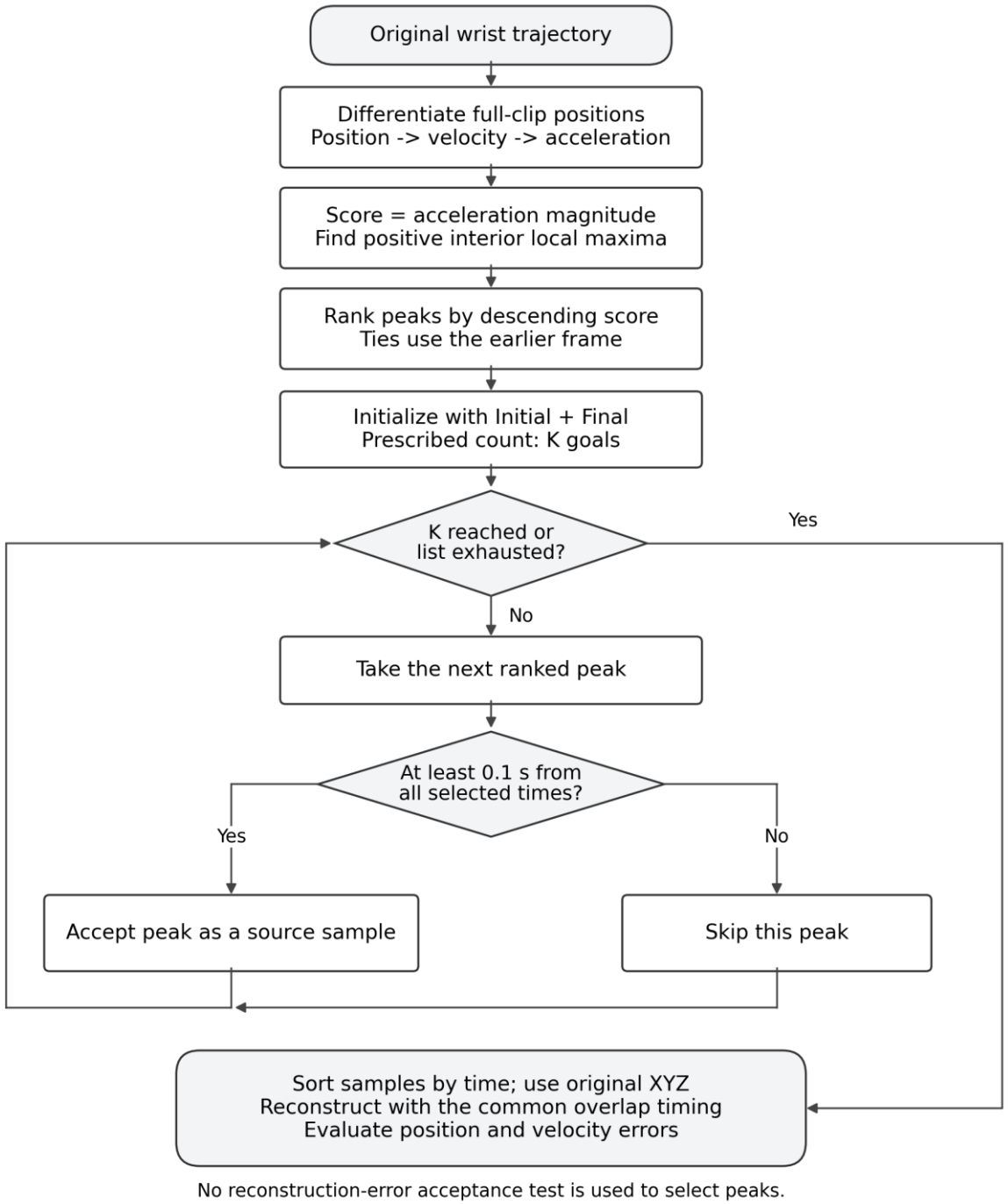
1. Compute the score and candidate peaks once from the original motion.
2. Rank candidates by descending score, breaking ties with the earlier frame.
3. Starting from the endpoints, accept the next candidate at least 0.1 seconds from all selected samples.
4. Continue until the requested number of goals is reached or no eligible peak remains. Smaller goal budgets use prefixes of this same insertion order.
5. Sort the selected samples in time, use their original positions as goals, reconstruct with the common timing rule, and evaluate.

No reconstruction error is used to rank, accept, or reject goals in this method. That separation is deliberate: otherwise it would partly become the error-driven method. If candidates run out, arbitrary non-peak points are not used to fill the budget.

The unsmoothed data contained 1,945 local candidate peaks and supplied enough eligible candidates for 102 goals. I inspected the candidates before the full comparison. They occurred near some visible changes, but also in relatively steady sections. The largest score, approximately 144.35 m/s^2 at 31.441667 seconds, coincided with a short position jump and return. Its physical origin has not been independently established.

I retained the raw version for this first comparison instead of adjusting smoothing after seeing its reconstruction error. This preserves a clear account of the original idea and its limitations. It does not imply that smoothing would be inappropriate; any smoothed version would need its own stated settings and separate evaluation against the same original reference.

5.1 Flowchart of acceleration based selection



Flowchart 2. Raw acceleration selection for a prescribed goal count K. Positive local peaks are ranked once, and accepted peaks must satisfy the 0.1 s separation from every selected source time. No smoothing or reconstruction-error acceptance test is used. The initial position is excluded from K; the final position is included.

6 What the candidate inspection showed

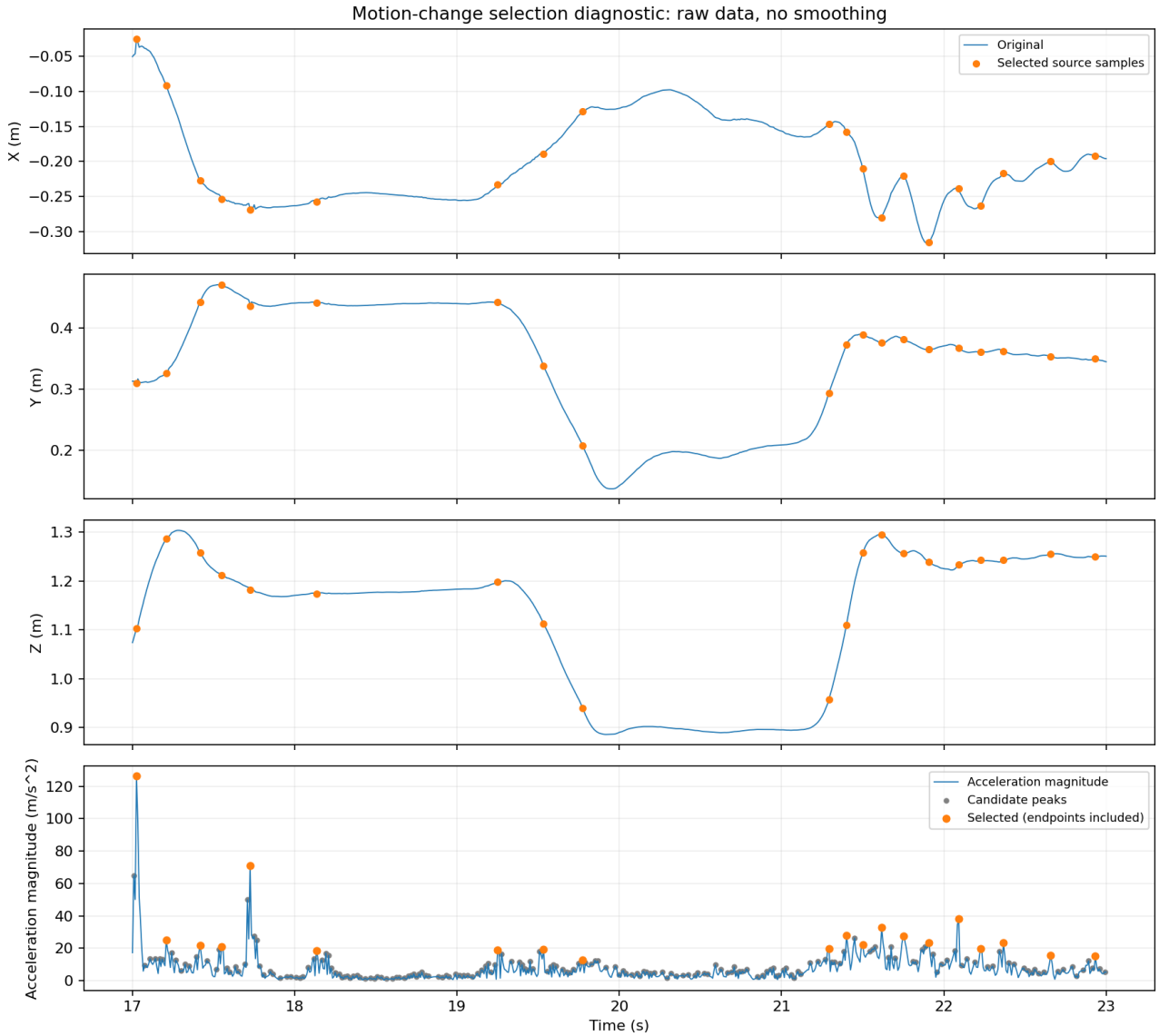


Figure 1. Candidate inspection over 17-23 seconds. The top three panels show original XYZ positions and the selected source samples in orange. The bottom panel shows raw acceleration magnitude, candidate peaks in gray, and selected samples in orange. This is a view of the full-clip 102-goal selection, not selection performed only within this window.

Near 21.5-22 seconds, four selected samples capture parts of the rapid changes; the sample at 21.908333 seconds is close to the X minimum at 21.900000 seconds. However, the interval 19.8-21.0 seconds contains no selected sample even though X, Y, and Z vary by about 5.86, 7.19, and 3.39 cm, respectively. In the relatively steady 8-12 second interval outside this view, seven samples are selected while each coordinate spans roughly 1-3 cm.

These observations show what the score prioritizes; they do not establish which points are useful until reconstruction is evaluated. In particular, a high acceleration score is not equivalent to a large reconstruction error, and small rapid variations cannot be labeled noise solely from these figures.

7 Comparison at equal goal counts

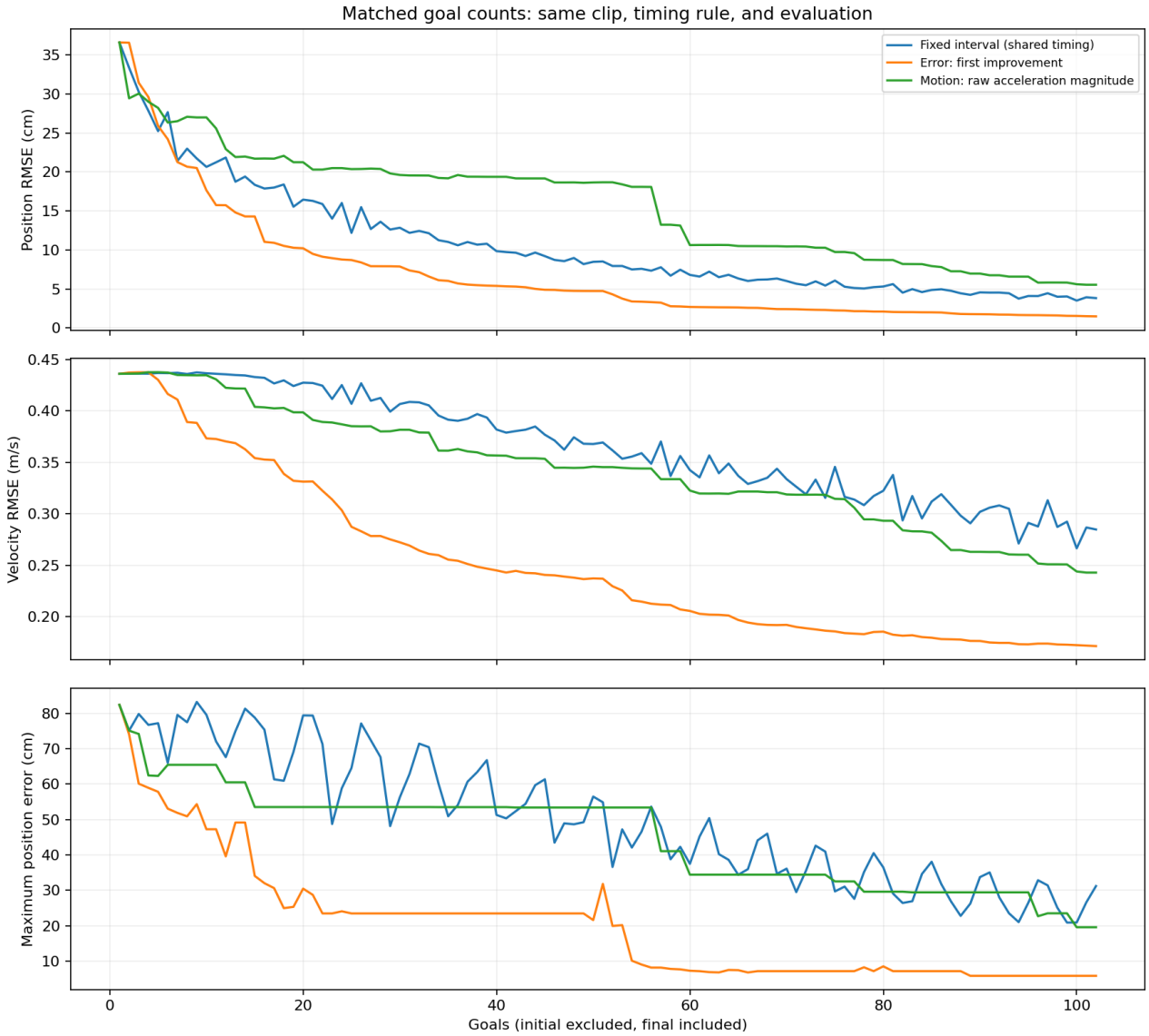


Figure 2. Position RMSE, velocity-vector RMSE, and maximum position error at each integer goal count from 1 to 102. All methods use the same clip, sampled goal positions, minimum source-time gap, timing rule, and evaluation. The error-driven curve contains accepted states; rejected trials are logged separately.

Method at 102 goals	Position RMSE (cm)	Velocity RMSE (m/s)	Maximum position error (cm)
Fixed interval	3.817	0.28461	31.204
Error driven first improvement	1.478	0.17128	5.846
Raw acceleration magnitude	5.531	0.24271	19.567

8 Interpreting the comparison

At 102 goals, the revised error-driven method has the lowest values of all three metrics in the main comparison. This supports using reconstruction feedback to allocate goals for this clip, but it does not establish global optimality or superiority on other motions. Position RMSE decreases by construction; the same is not true of velocity RMSE or the maximum position error.

The motion method gives higher position RMSE than uniform sampling, but lower velocity RMSE and lower maximum position error. A feature-based allocation can therefore improve some aspects without minimizing the overall positional discrepancy. In Figure 7, the 102-goal curve follows several sharp changes but misses the lower Y and Z behavior around 20-21 seconds. The exact contributions of candidate placement, raw derivative fluctuations, and the shared timing heuristic have not been isolated.

The comparison holds representation size equal, not computation or objective. The error-driven method explicitly tests position RMSE and makes thousands of reconstruction trials; the motion method ranks source features without such acceptance tests. The former's lower position RMSE comes with additional computation. No wall-clock performance benchmark or real-time claim is made.

9 Effect of goal count

I considered stopping when reconstruction error became small enough, or when adding another goal produced only a small improvement. However, I could not justify a particular threshold for this experiment without a specified acceptable reconstruction accuracy or a cost for additional goals. I therefore did not introduce a threshold as if it identified the correct number of goals. This does not mean that a suitable criterion cannot be defined for an application.

Instead, I extended the analysis of the existing results at 10, 20, 40, 60, 80, and 102 goals. These are representative counts spanning the tested range, not proposed optimal counts. The following table summarizes full-clip position and velocity RMSE. Sections 9.1-9.3 then compare different counts within each method, with separate position and velocity figures. Each figure shows 10, 20, and 40 goals on the left, and 60, 80, and 102 goals on the right, over the same 17-23 second interval. Axis limits are shared across methods and counts. All goals were selected over the whole clip.

Each table entry below is position RMSE in cm / velocity RMSE in m/s. K excludes the initial position and includes the final position.

K	Fixed interval Position / velocity	Error driven Position / velocity	Acceleration Position / velocity
10	20.639 / 0.43644	17.635 / 0.37316	26.981 / 0.43450
20	16.441 / 0.42728	10.207 / 0.33110	21.234 / 0.39842
40	9.838 / 0.38174	5.384 / 0.24483	19.366 / 0.35650
60	6.795 / 0.34231	2.685 / 0.20545	10.615 / 0.32244
80	5.304 / 0.32224	2.095 / 0.18533	8.704 / 0.29308
102	3.817 / 0.28461	1.478 / 0.17128	5.531 / 0.24271

At these six counts, the error-driven method has the lowest full-clip position and velocity RMSE among the three methods. This ranking does not necessarily hold in every time interval. Its position RMSE decreases from 5.384 cm at 40 goals to 2.685 cm at 60, then to 2.095 cm at 80 and 1.478 cm at 102. These changes describe the tradeoff but do not by themselves establish an acceptable stopping point.

Other metrics can move differently: from 60 to 80 goals, the error-driven maximum position error rises from 7.281 cm to 8.520 cm even while its position and velocity RMSE decrease. Thus, a stopping condition based on one metric would not establish that every aspect of the reconstruction is satisfactory.

9.1 Fixed interval across goal counts

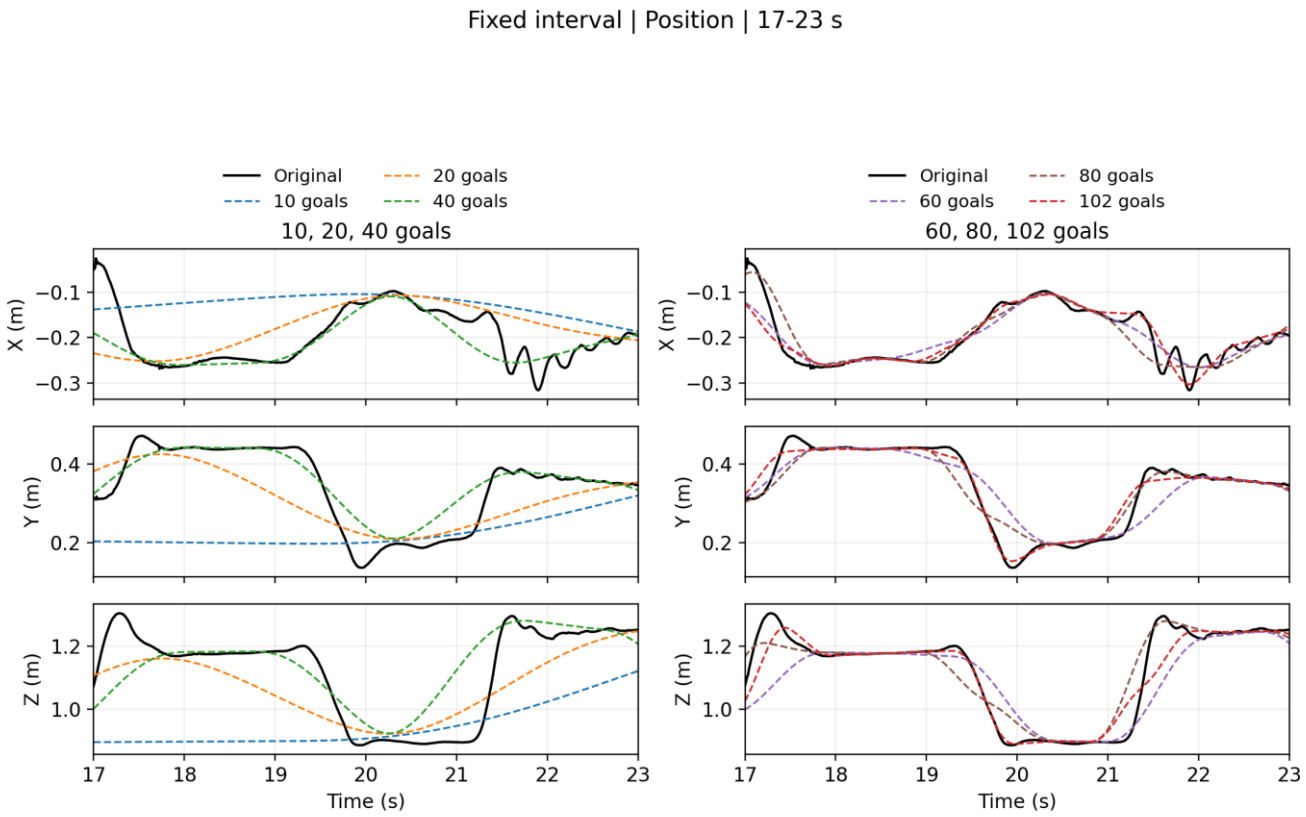


Figure 3. Fixed interval: XYZ position at different goal counts over 17-23 seconds. Left: 10, 20, and 40 goals. Right: 60, 80, and 102 goals. Black is the original trajectory; the legends identify each goal count. Axis limits are identical across methods for each coordinate and quantity. All goal selection and reconstruction use the full clip.

Increasing the count progressively describes more of the Y and Z descent and return near 20-21.5 seconds, but even the 102-goal curve rounds the rapid transitions. The table below reports full-clip metrics; this figure displays only the selected window.

Full-clip metrics at each count; these are not restricted to the displayed window.

Goals	Position RMSE (cm)	Velocity RMSE (m/s)	Maximum position error (cm)
10	20.639	0.43644	79.566
20	16.441	0.42728	79.397
40	9.838	0.38174	51.280
60	6.795	0.34231	37.474
80	5.304	0.32224	36.468
102	3.817	0.28461	31.204

Fixed interval velocity comparison

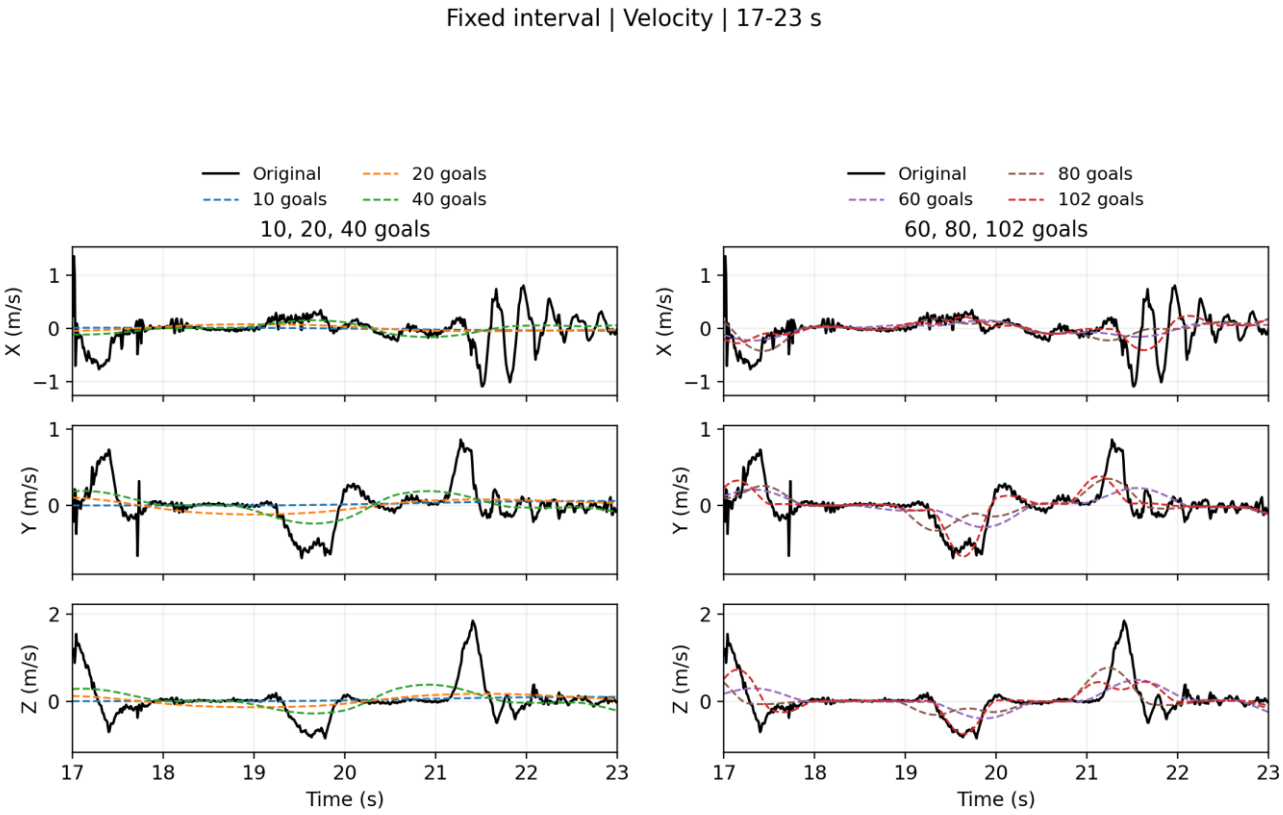


Figure 4. Fixed interval: XYZ velocity at different goal counts over 17-23 seconds. Left: 10, 20, and 40 goals. Right: 60, 80, and 102 goals. Black is the original trajectory; the legends identify each goal count. Axis limits are identical across methods for each coordinate and quantity. Velocities are differentiated over the full clip before cropping.

The velocity comparison makes the missed dynamics more apparent than position alone. Low counts produce slow, broad changes; higher counts resolve more structure but still miss several short peaks, especially in X near 21.5-22 seconds. Full-clip velocity RMSE decreases from 0.43644 m/s at 10 goals to 0.28461 m/s at 102; the displayed window still has 0.43708 m/s at 102. The fixed samples are recomputed at each count, so these curves do not result from retaining every earlier sample.

9.2 Error driven first improvement across goal counts

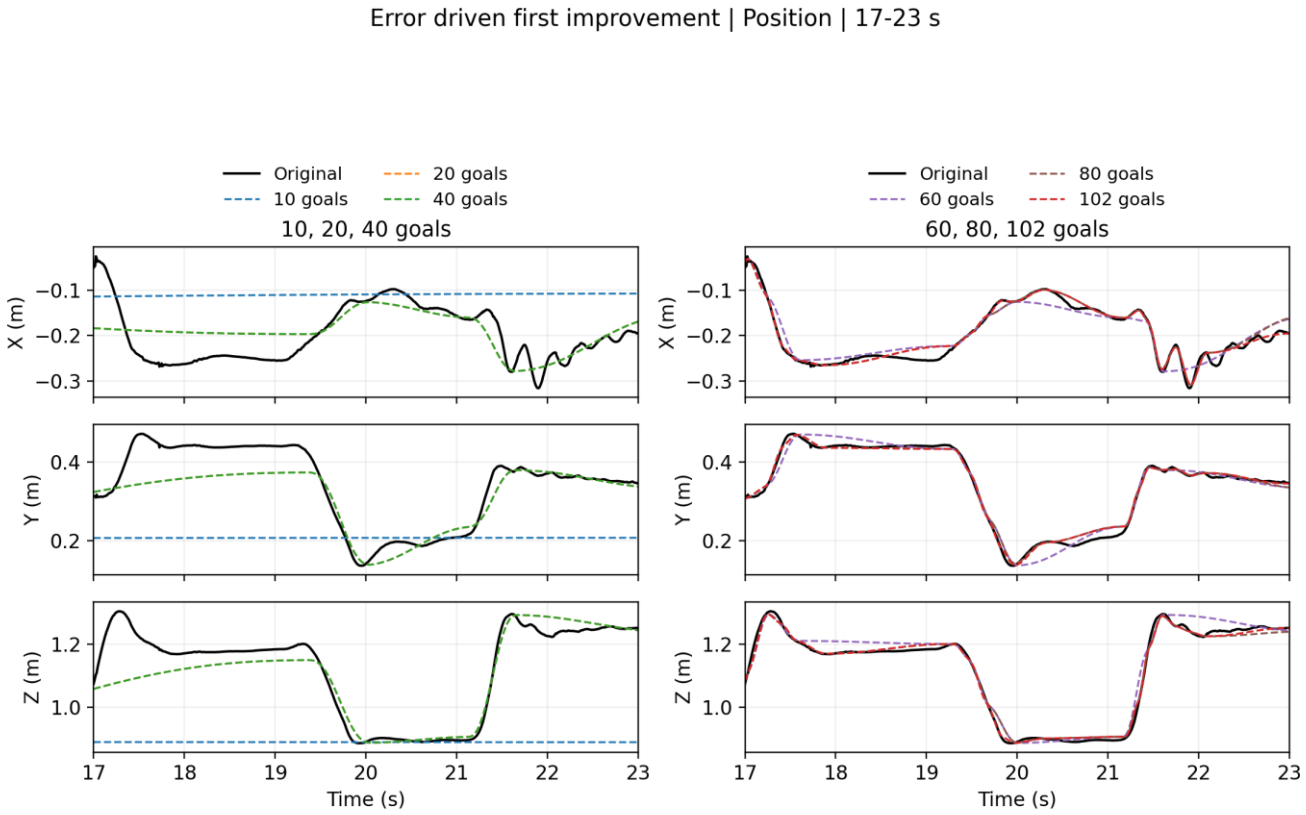


Figure 5. Error driven first improvement: XYZ position at different goal counts over 17-23 seconds. Left: 10, 20, and 40 goals. Right: 60, 80, and 102 goals. Black is the original trajectory; the legends identify each goal count. Axis limits are identical across methods for each coordinate and quantity. All goal selection and reconstruction use the full clip.

The 20- and 40-goal curves nearly coincide in this window, although full-clip position RMSE improves from 10.207 to 5.384 cm. Their window RMSE remains approximately 9.216 cm. From 60 to 102 goals, more of the local descent, return, and short X minima are captured. A global improvement can therefore be concentrated outside the displayed interval.

Full-clip metrics at each count; these are not restricted to the displayed window.

Goals	Position RMSE (cm)	Velocity RMSE (m/s)	Maximum position error (cm)
10	17.635	0.37316	47.245
20	10.207	0.33110	30.470
40	5.384	0.24483	23.467
60	2.685	0.20545	7.281
80	2.095	0.18533	8.520
102	1.478	0.17128	5.846

Error driven first improvement velocity comparison

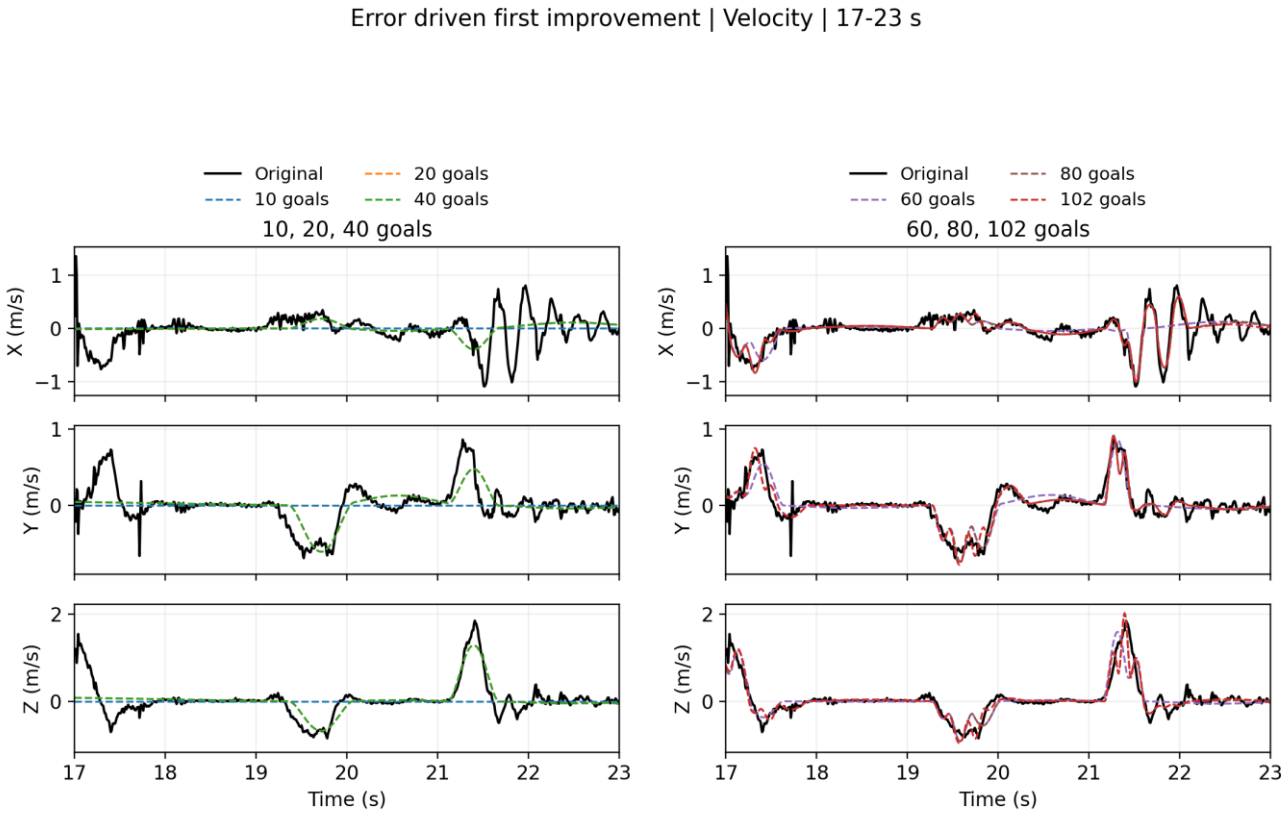


Figure 6. Error driven first improvement: XYZ velocity at different goal counts over 17-23 seconds. Left: 10, 20, and 40 goals. Right: 60, 80, and 102 goals. Black is the original trajectory; the legends identify each goal count. Axis limits are identical across methods for each coordinate and quantity. Velocities are differentiated over the full clip before cropping.

Velocity improves substantially between the low and high count groups, but short peaks still differ in amplitude or timing. Window velocity RMSE falls from 0.56351 m/s at 10 goals to 0.23620 m/s at 102. These curves also explain why similar position traces do not guarantee identical velocities. Full-clip maximum position error rises from 7.281 cm at 60 goals to 8.520 cm at 80 despite lower position and velocity RMSE. At 10 goals, this method has the lowest full-clip position RMSE but the highest window position RMSE among the three methods (33.856 cm).

9.3 Raw acceleration magnitude across goal counts

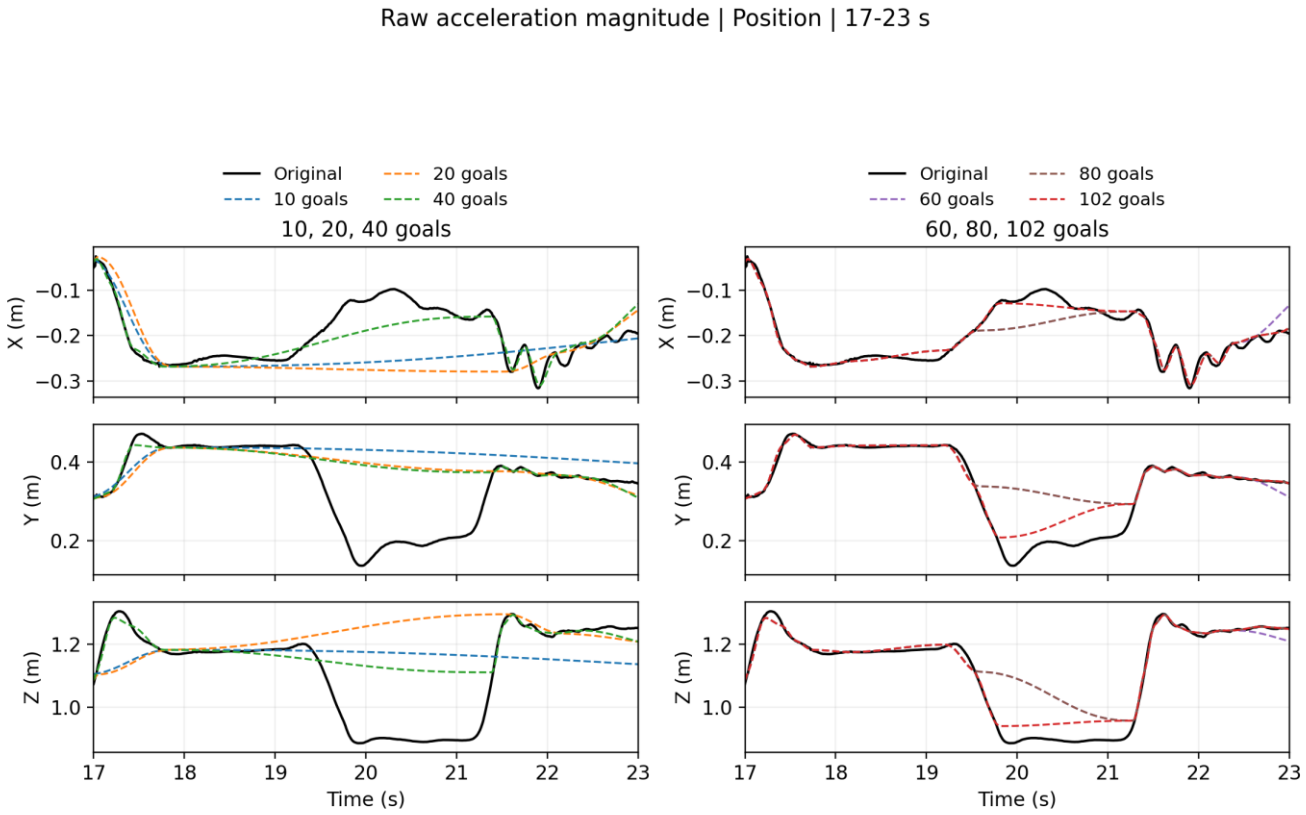


Figure 7. Raw acceleration magnitude: XYZ position at different goal counts over 17-23 seconds. Left: 10, 20, and 40 goals. Right: 60, 80, and 102 goals. Black is the original trajectory; the legends identify each goal count. Axis limits are identical across methods for each coordinate and quantity. All goal selection and reconstruction use the full clip.

With more goals, the reconstruction captures several rapid X changes near 21.5-22 seconds. It nevertheless misses the sustained low Y and Z positions around 20-21 seconds, including at 102 goals. Full-clip position RMSE falls from 26.981 cm at 10 goals to 5.531 cm at 102. Locally, it can worsen: window RMSE rises from 21.322 cm at 10 goals to 24.462 cm at 20.

Full-clip metrics at each count; these are not restricted to the displayed window.

Goals	Position RMSE (cm)	Velocity RMSE (m/s)	Maximum position error (cm)
10	26.981	0.43450	65.437
20	21.234	0.39842	53.515
40	19.366	0.35650	53.491
60	10.615	0.32244	34.413
80	8.704	0.29308	29.597
102	5.531	0.24271	19.567

Raw acceleration magnitude velocity comparison

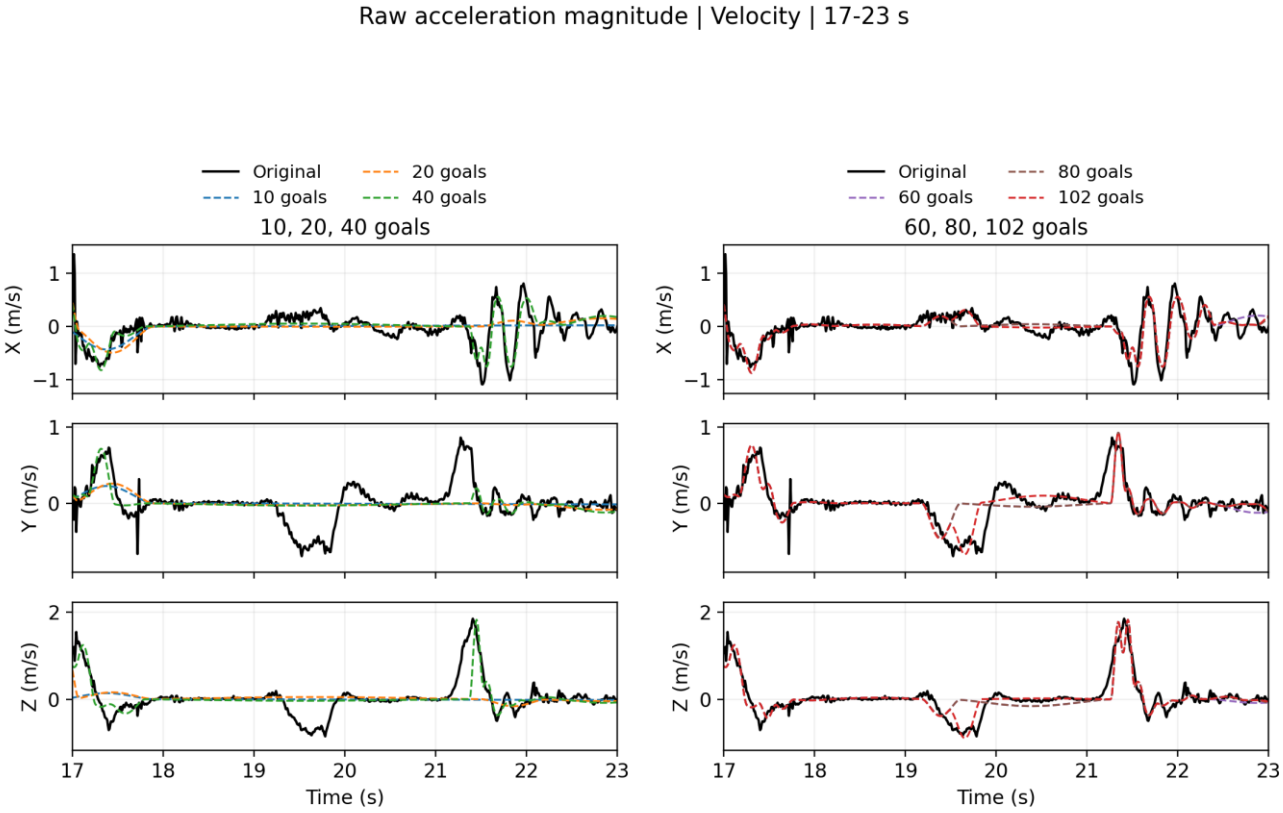


Figure 8. Raw acceleration magnitude: XYZ velocity at different goal counts over 17-23 seconds. Left: 10, 20, and 40 goals. Right: 60, 80, and 102 goals. Black is the original trajectory; the legends identify each goal count. Axis limits are identical across methods for each coordinate and quantity. Velocities are differentiated over the full clip before cropping.

The higher-count curves capture several rapid velocity changes, but their peak amplitudes and timings remain imperfect. Full-clip velocity RMSE decreases from 0.43450 m/s at 10 goals to 0.24271 m/s at 102. Some intermediate additions produce little benefit in this window: window position RMSE changes from 10.033 cm at 60 goals to 9.955 cm at 80. Acceleration ranking does not directly minimize either reconstruction metric, and these observations do not justify a unique final goal count.

10 Comparison details and remaining limitations

For each K, the fixed baseline selects exactly K+1 uniformly distributed source-frame indices across the clip, rounded to the nearest frame and including both endpoints. It is recomputed independently at each count. This differs from the original 0.5-second baseline, which appended a shorter final interval: both sampling locations and the timing rule differ. The new results must not be interpreted as unchanged first-stage results. The original report and outputs are retained.

All three methods were evaluated at every count from 1 through 102, giving 306 states without missing budgets. The error-driven cached states were reconstructed again and their positions and metrics checked against the recorded run. No comparison parameters were retuned after observing the three-method results. Twenty-five numerical and selection checks passed, including known derivatives, hand-calculated metrics, rollback, reconsideration, spacing, candidate exhaustion, and exact count handling.

The comparison measures reconstruction on the same clip used for selection; it is not held-out prediction. Goal positions are not fitted, and activation/completion times follow a neighbor-dependent heuristic rather than being independently fitted to motion events. The two adaptive methods determine an ordering of goals from the data, but this run uses a prescribed maximum budget of 102. It does not identify a unique appropriate number of goals or an application-based accuracy threshold. The error method reached the budget, so its no-improvement stopping rule was tested on small examples but did not determine the final count in this clip.

The 0.1-second minimum gap, 0.25 extension fraction, and raw acceleration score remain initial settings. Raw differentiation can emphasize small fluctuations, but their origin and the effect of smoothing have not been established. These limitations leave open whether improved timing, a different motion-change score, or fitted positions would change the comparison.

11 Reproduction and reference

From the repository root, use the working WSL Python environment from the first experiment:

```
python3 -m unittest discover -s tests -v
python3 -m src.followup.error_insertion
python3 -m src.followup.accepted_error_insertion
python3 -m src.followup.motion_change
python3 -m src.followup.compare_methods
python3 -m src.followup.analyze_goal_counts
```

The input is results/wrist_trajectory.npz. Run the comparison after accepted insertion. The analysis verifies 18 saved trajectories and writes figures and full-clip/window metrics to results/followup/goal_count_analysis/ without rerunning selection. Dependency versions and setup notes are in README.md; a clean installation has not been tested. Code and data access: https://github.com/Hikaru-shih/minimum-jerk-motion-approximation.

H. Sato, H. Mitake, and S. Hasegawa. "Artist-Directable Motion Generation Using Overlapping Minimum Jerk Trajectories for Interactive Embodied Social Agent Adapting to Dynamic Environments." Computer Animation and Virtual Worlds, 36, e70075, 2025. https://doi.org/10.1002/cav.70075.